

**Network Security (CS3403)
Project Report**

A Secure and Encrypted Web-Based Voting System

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1. Abstract

This project implements a secure web-based voting system that ensures confidentiality, integrity, and authenticity of votes. The system utilizes modern cryptographic techniques for vote encryption, user authentication, and secure data transmission. Built with a React frontend and Node.js backend, PostgreSQL, it features role-based access control, real-time vote tracking, and encrypted storage of voting data, making it suitable for small to medium-scale electronic voting scenarios.

2. Introduction

Project Background and Relevance

Electronic voting systems are becoming increasingly important in modern democracy, but they must address critical security concerns. This project demonstrates a practical implementation of secure e-voting principles using web technologies while maintaining voter privacy and vote integrity.

Objectives:

- Implement secure user authentication and authorization
- Ensure vote confidentiality through encryption
- Prevent double voting
- Provide real-time vote tracking for administrators
- Maintain audit trail of voting activities
- Demonstrate secure key management practices

3. System Overview

System Architecture:

Frontend: React-based single-page application(SPA)

- - Provides responsive user interface
- - Handles client-side encryption
- - Manages user sessions
- - Real-time updates via WebSocket
- - Form validation and error handling

Backend: Node.js/Express REST API

- - Handles API requests
- - Processes votes
- - Manages authentication
- - Database operations
- - WebSocket server management

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Database: PostgreSQL for persistent storage

- - Secure data storage
- - Relationship management
- - Data integrity
- - Audit trail maintenance

WebSocket server for real-time updates

- - Enables real-time bidirectional communication
- - More efficient than polling for live updates
- - Maintains persistent connection
- - Low latency for instant vote updates

JWT-based authentication

- Uses JSON Web Tokens for secure user authentication.
- Ensures stateless session management with role-based access control.
- Enhances scalability and security.

Cryptographic services for vote encryption/decryption

- Employs asymmetric encryption to secure votes.
- Ensures confidentiality and integrity of votes during transmission and storage.
- Allows authorized parties to decrypt and verify votes securely.

Data Flow:

1. **User Registration/Authentication:** Users register with unique credentials, and cryptographic key pairs are generated for secure voting.
2. **Encrypted Vote Submission:** Votes are encrypted on the client side before submission to the server.
3. **Real-Time Vote Verification:** Admins verify votes in real time through the dashboard.
4. **Secure Vote Storage and Counting:** Encrypted votes are stored in the database, ensuring confidentiality and integrity.

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4. Design

Major Components:

a) Authentication System

- **JWT-Based Token Management:**
 - Utilizes JSON Web Tokens (JWT) for secure authentication.
 - Tokens are generated upon successful login and contain user claims.
 - Ensures stateless authentication, reducing server load.
- **Role-Based Access Control (Voter/Admin):**
 - Assigns roles to users (voter or admin) to control access to features.
 - Admins have access to sensitive functions like vote verification and user management.
- **Password Hashing Using Bcrypt:**
 - Stores passwords securely by hashing them using bcrypt.
 - Protects against unauthorized access even if the database is compromised.

b) Encryption Module

- **Asymmetric Encryption for Votes Confidentiality:**
 - Uses public-key cryptography to encrypt votes.
 - Ensures that only authorized parties with the private key can decrypt votes.
- **Secure Key Pair Generation During Registration:**
 - Generates a unique public/private key pair for each user during registration.
 - The public key is used for encryption, while the private key is kept secure for decryption.
- **Client-Side Encryption:**
 - Votes are encrypted on the client side before being sent to the server.
 - Enhances security by ensuring that votes are never transmitted in plain text.

c) Vote Management

- **Encrypted Vote Storage:**
 - Stores votes in an encrypted form in the database.
 - Protects against unauthorized access to voting data.
- **Double-Voting Prevention:**
 - Tracks user voting status in real-time to prevent multiple votes from the same user.
 - Ensures the integrity of the voting process.
- **Real-Time Vote Counting:**
 - Updates vote counts instantly as votes are cast.

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- Provides administrators with live insights into election progress.

d) Admin Dashboard

- **Vote Verification Interface:**
 - Allows administrators to verify the authenticity of votes.
 - Ensures that votes are legitimate and have not been tampered with.
- **Real-Time Statistics:**
 - Displays live statistics on voting progress and results.
 - Helps administrators monitor the election in real-time.
- **User Management:**
 - Provides tools for managing user accounts, including registration and role assignment.
 - Enables administrators to oversee user activity and ensure system security.

5. Security Features

- 1. Password Hashing Using Bcrypt:**
 - Purpose: Securely stores user passwords by hashing them using bcrypt.
 - Benefits: Protects passwords from unauthorized access even if the database is compromised.
- 2. JWT for Secure Session Management:**
 - Purpose: Uses JSON Web Tokens to manage user sessions securely.
 - Benefits: Ensures stateless authentication, reducing server load and enhancing scalability.
- 3. Asymmetric Encryption for Vote Data:**
 - Purpose: Encrypts votes using public-key cryptography to ensure confidentiality.
 - Benefits: Only authorized parties with the private key can decrypt votes, maintaining voter anonymity and preventing tampering.
- 4. HTTPS for Secure Data Transmission:**
 - Purpose: Encrypts all data exchanged between the client and server.
 - Benefits: Protects against eavesdropping and interception of sensitive information during transmission.
- 5. Protection Against Double Voting:**
 - Purpose: Tracks user voting status in real-time to prevent multiple votes from the same user.
 - Benefits: Ensures the integrity of the voting process by preventing fraudulent activities.

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6. Role-Based Access Control:

- Purpose: Assigns roles to users (e.g., voter or admin) to control access to system features.
- Benefits: Restricts unauthorized access to sensitive functions like vote verification and user management.

7. Secure Key Storage:

- Purpose: Safely stores cryptographic keys used for encryption and decryption.
- Benefits: Protects against unauthorized access to keys, ensuring the security of encrypted data.

8. Input Validation and Sanitization:

- Purpose: Checks and cleans user input to prevent malicious data from entering the system.
- Benefits: Protects against common web vulnerabilities like SQL injection and cross-site scripting (XSS).

9. Protection Against Common Web Vulnerabilities:

- Purpose: Implements measures to defend against known web threats.
- Benefits: Enhances overall system security by reducing the risk of attacks and data breaches.

6. System Requirements

Platform Requirements:

1. Node.js v14+ Runtime:

- Purpose: Runs the backend server, handling API requests and database operations.
- Benefits: Ensures compatibility with modern Node.js features and security patches.

2. PostgreSQL v12+ Database:

- Purpose: Stores user data, votes, and other critical information securely.
- Benefits: Offers robust data management capabilities with high security and reliability.

3. Modern Web Browser with JavaScript Enabled:

- Purpose: Provides a responsive and interactive user interface for voters and administrators.
- Benefits: Ensures compatibility across different devices and browsers.

4. Windows/Linux/macOS Operating System:

- Purpose: Supports development and deployment across various platforms.
- Benefits: Offers flexibility in choosing the operating system for development and deployment.

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Network Requirements:

1. Stable Internet Connection:

- Purpose: Ensures reliable communication between clients and servers.
- Benefits: Prevents disruptions during voting and ensures real-time updates.

2. Support for WebSocket Connections:

- Purpose: Enables real-time communication for instant vote updates and live statistics.
- Benefits: Enhances user experience with immediate feedback and updates.

3. HTTP/HTTPS Protocol Support:

- Purpose: Secures data transmission between clients and servers.
- Benefits: Protects against eavesdropping and tampering with sensitive information.

7. Open-source Libraries and Tools

Backend:

- **Express.js v4.18.2:** Web application framework
- **Sequelize v6.35.2:** ORM for database operations
- **bcryptjs v2.4.3:** Password hashing
- **jsonwebtoken v9.0.0:** JWT implementation
- **crypto-js v4.1.1:** Cryptographic operations
- **ws v8.18.1:** WebSocket server

Frontend:

- **React v18.2.0:** UI framework
- **React Router v6.8.1:** Client-side routing
- **TailwindCSS v3.2.4:** Styling
- **crypto-js v4.1.1:** Client-side encryption

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8. Implementation and Testing

Testing Methodology:

- Unit testing of cryptographic functions
- Integration testing of API endpoints
- End-to-end testing of voting process
- Security testing for authentication
- Load testing for concurrent users
- Cross-browser compatibility testing

Test Scenarios:

- User registration and login
- Vote submission and encryption
- Admin dashboard functionality
- Real-time update verification
- Error handling and validation

9. Results

Implemented Features:

- Secure user authentication
- Encrypted vote submission
- Real-time vote tracking
- Admin dashboard with vote verification
- Prevention of double voting
- Audit trail of voting activities

Future Extensions:

- Blockchain integration for immutable vote records
- Multi-factor authentication
- Advanced analytics dashboard
- Support for multiple simultaneous elections

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- Mobile application support
- Offline voting capability

10. Conclusion

The project successfully demonstrates a practical implementation of a secure electronic voting system. Key learnings include:

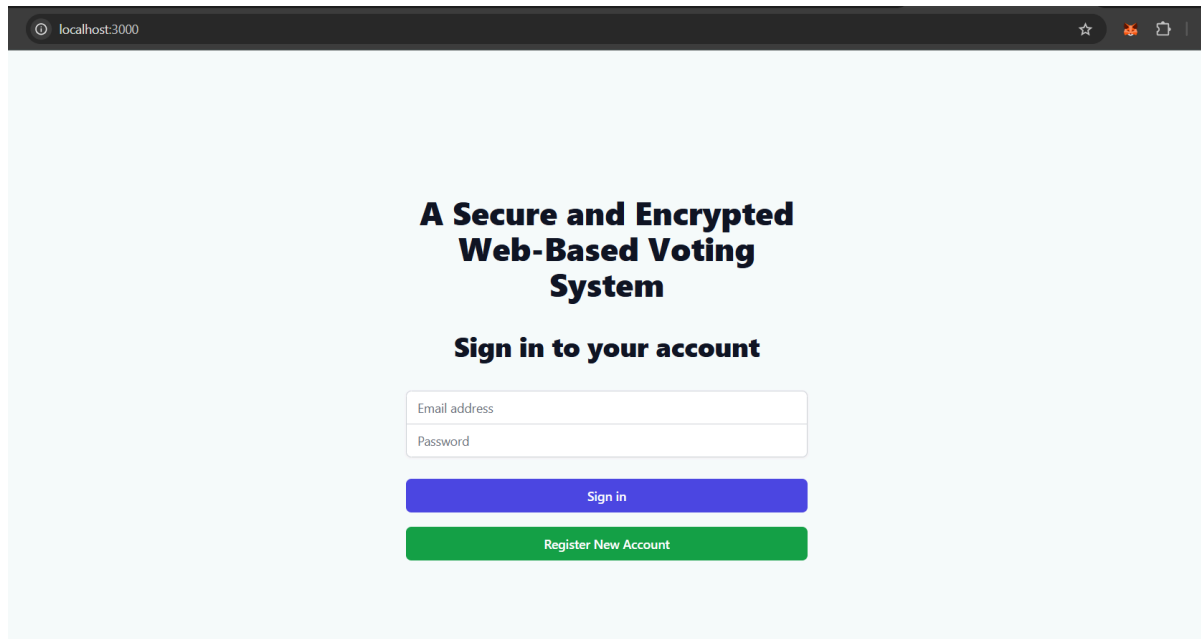
- Implementation of cryptographic protocols in web applications
- Real-time data synchronization techniques
- Secure user authentication practices
- Database design for sensitive information
- Frontend-backend security integration

The system provides a foundation for secure electronic voting while maintaining user privacy and vote integrity. The modular architecture allows for future enhancements and scaling to support larger elections.

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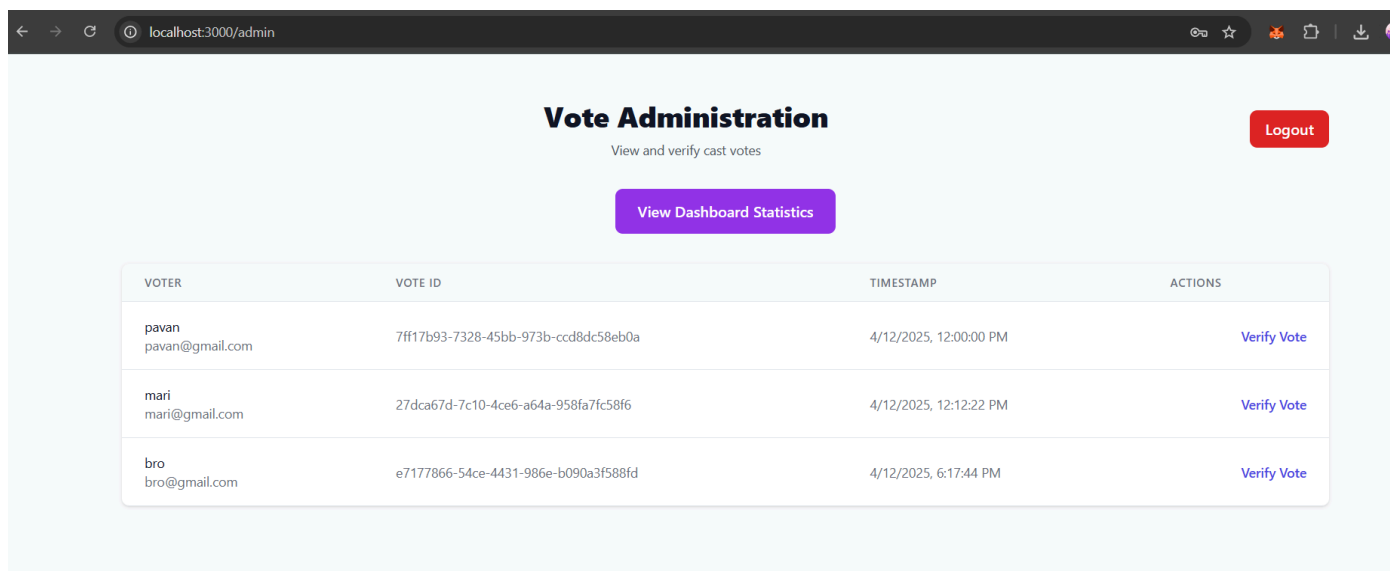
Output of the Project

Sign in Page



A screenshot of a web browser showing the sign-in page for a voting system. The browser address bar shows 'localhost:3000'. The page has a light blue background. In the center, there is a heading 'A Secure and Encrypted Web-Based Voting System' followed by 'Sign in to your account'. Below this, there are two input fields: 'Email address' and 'Password'. Under the input fields, there are two buttons: a blue 'Sign in' button and a green 'Register New Account' button.

Admin Login

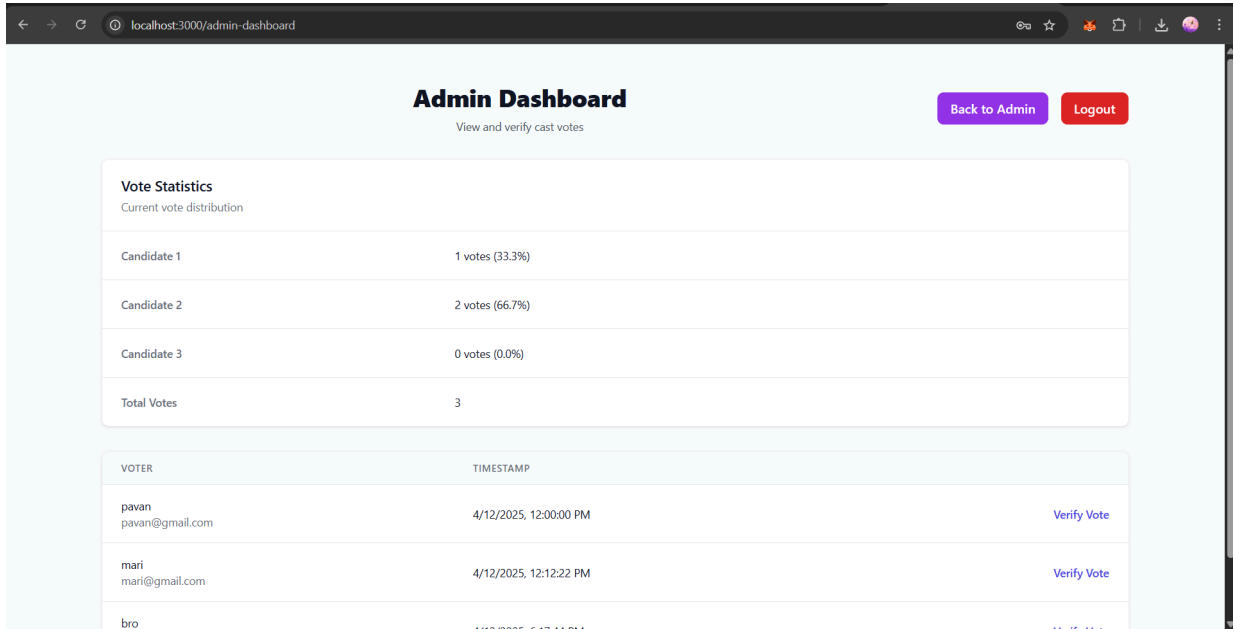


A screenshot of a web browser showing the admin login page. The browser address bar shows 'localhost:3000/admin'. The page has a light blue background. At the top, there is a heading 'Vote Administration' with a subtitle 'View and verify cast votes'. To the right of the heading is a red 'Logout' button. Below the heading, there is a purple button labeled 'View Dashboard Statistics'. At the bottom, there is a table with four columns: 'VOTER', 'VOTE ID', 'TIMESTAMP', and 'ACTIONS'.

VOTER	VOTE ID	TIMESTAMP	ACTIONS
pavan pavan@gmail.com	7ff17b93-7328-45bb-973b-ccd8dc58eb0a	4/12/2025, 12:00:00 PM	Verify Vote
mari mari@gmail.com	27dca67d-7c10-4ce6-a64a-958fa7fc58f6	4/12/2025, 12:12:22 PM	Verify Vote
bro bro@gmail.com	e7177866-54ce-4431-986e-b090a3f588fd	4/12/2025, 6:17:44 PM	Verify Vote

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Admin dashboard

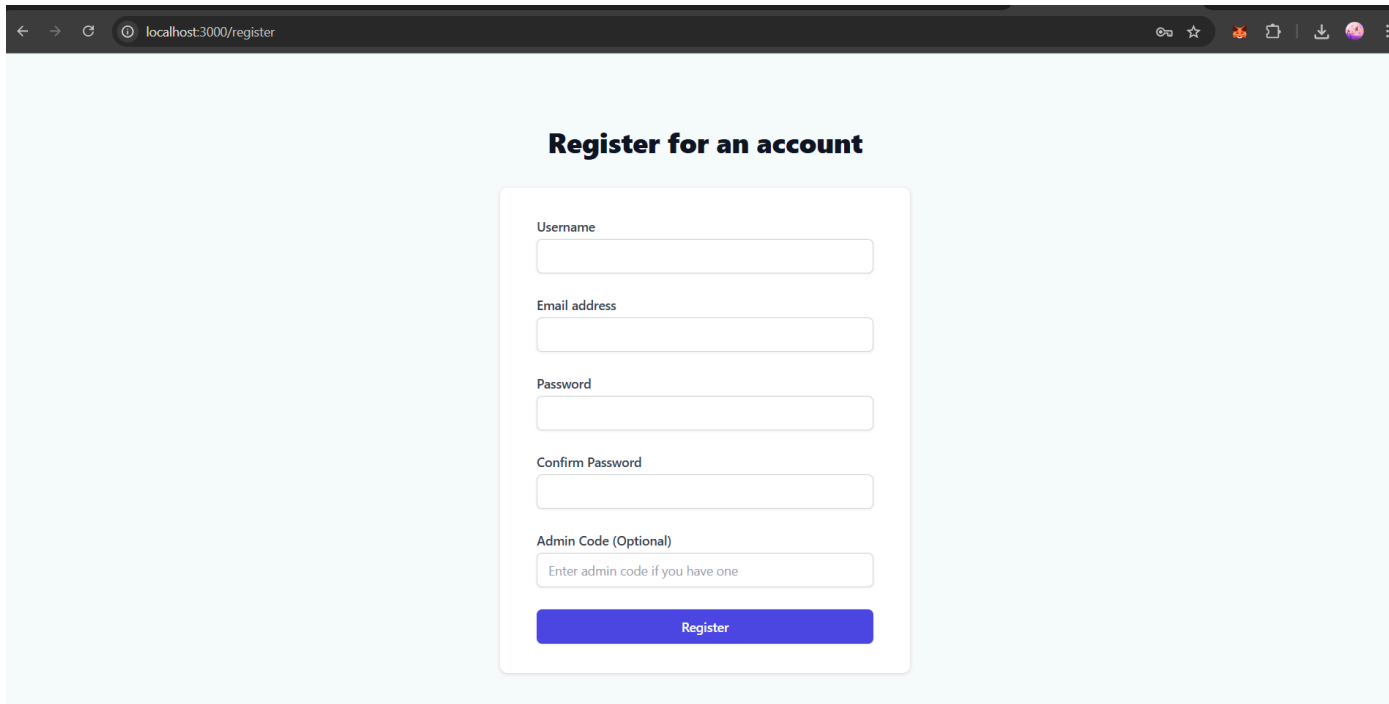


The screenshot shows the Admin Dashboard interface. At the top, there's a header with the title "Admin Dashboard" and a subtitle "View and verify cast votes". On the right, there are two buttons: "Back to Admin" (purple) and "Logout" (red). Below the header, there's a "Vote Statistics" section with the subtitle "Current vote distribution". It contains a table with the following data:

Candidate	Votes	Percentage
Candidate 1	1 votes	33.3%
Candidate 2	2 votes	66.7%
Candidate 3	0 votes	0.0%
Total Votes	3	

Below the statistics, there's a table of voters with columns "VOTER" and "TIMESTAMP". It lists three voters: pavan, mari, and bro, each with their email address and the timestamp of their vote. A "Verify Vote" link is provided for each voter.

New User or Admin to register



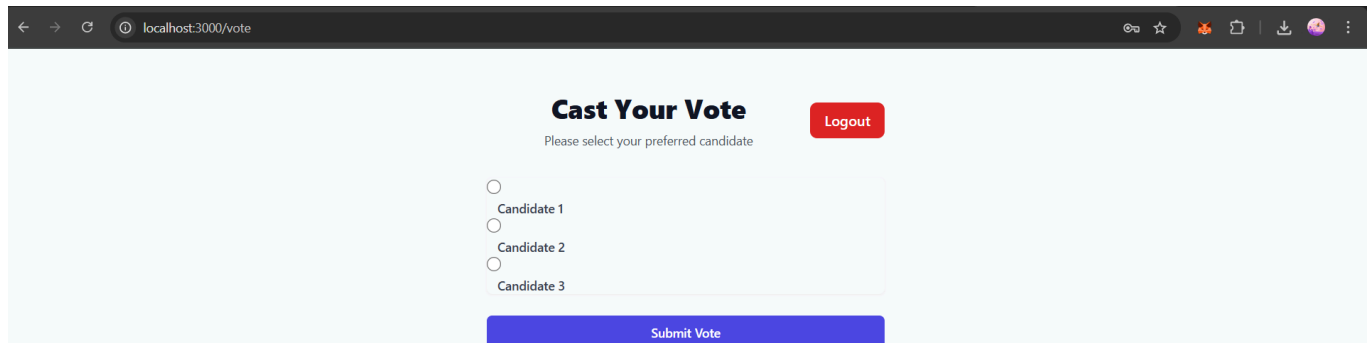
The screenshot shows the "Register for an account" form. It has a title "Register for an account" and a subtitle "Register for an account". The form contains the following fields:

- Username**: A text input field.
- Email address**: A text input field.
- Password**: A text input field.
- Confirm Password**: A text input field.
- Admin Code (Optional)**: A text input field with a placeholder "Enter admin code if you have one".

At the bottom of the form, there is a blue "Register" button.

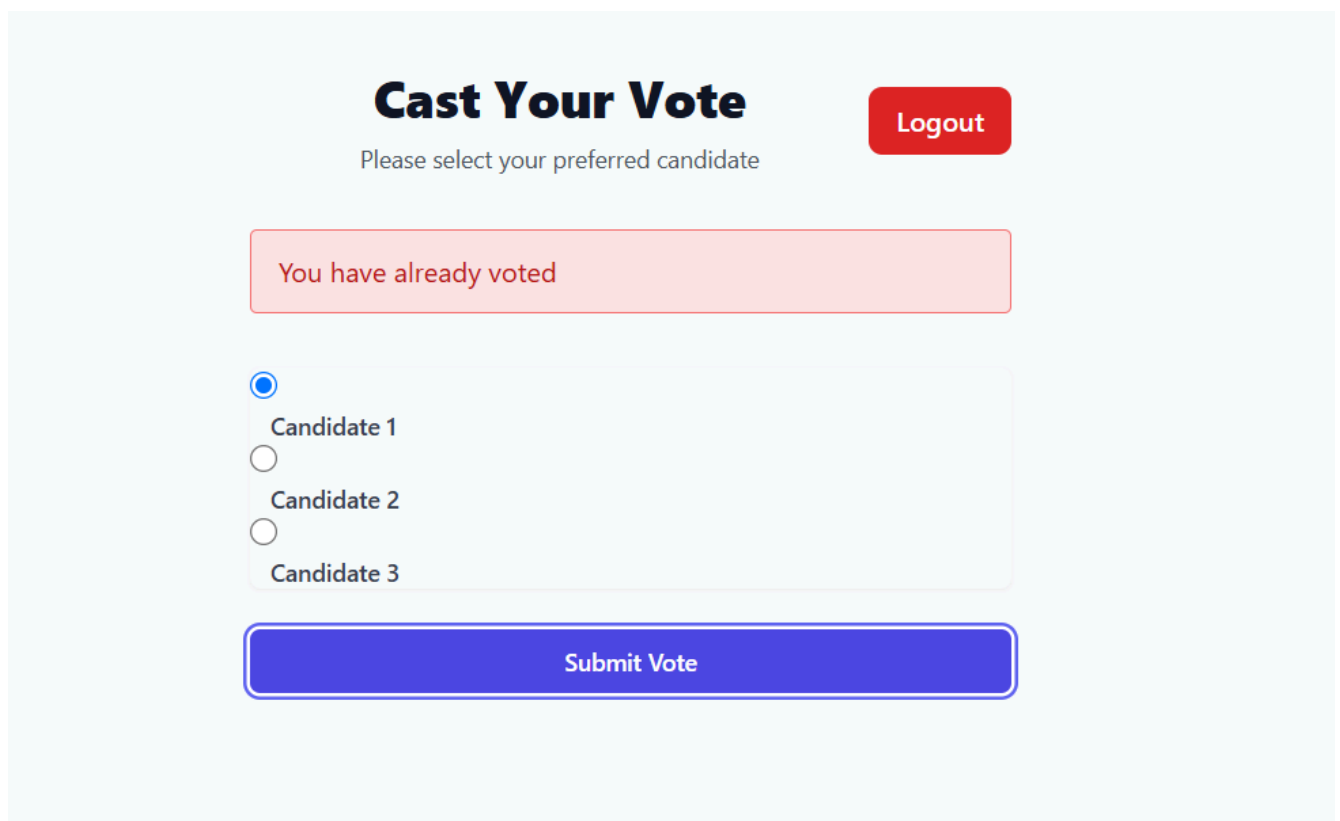
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To Cast Vote



The screenshot shows a web browser window with the address bar displaying 'localhost:3000/vote'. The page has a light blue background. At the top center, the text 'Cast Your Vote' is displayed in bold, with the instruction 'Please select your preferred candidate' below it. To the right of this text is a red 'Logout' button. Below the instruction, there is a white box containing three radio button options: 'Candidate 1', 'Candidate 2', and 'Candidate 3'. At the bottom center of the page is a large blue 'Submit Vote' button.

If I Cast Vote Again (as a existing user)



The screenshot shows the same 'Cast Your Vote' web application interface. In the center, there is a red-bordered box with the text 'You have already voted' in red. Below this box, the three radio button options 'Candidate 1', 'Candidate 2', and 'Candidate 3' are visible. The 'Candidate 1' radio button is selected, indicated by a blue dot. The 'Submit Vote' button at the bottom is still present and highlighted with a blue border.

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If the credentials are wrong and if not registered

A Secure and Encrypted Web-Based Voting System

Sign in to your account

Invalid email or password

fun@gmail.com

....

Sign in

Register New Account

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If the user already exists

Register for an account

User already exists

Username

pavan

Email address

pavan@gmail.com

Password

....

Confirm Password

....

Admin Code (Optional)

Enter admin code if you have one

Register

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User Table in DataBase (Postgres)

pgAdmin 4

Welcome public.Users/voting/postgres@PostgreSQL 17 x public.Votes/votin... x

public.Users/voting/postgres@PostgreSQL 17

Query Query History

```
1 SELECT * FROM public."Users"
2 ORDER BY id ASC
```

Scratch Pad x

Data Output Messages Notifications

Showing rows: 1 to 5 Page No: 1 of 1

	id [PK] uuid	username character varying (255)	email character varying (255)	password character varying (255)	role 'enum_Users_role'	hasVoted boolean	publicKey character varying (255)	createdAt timestamp with time zone	updatedAt timestamp with time zone
1	16bfd7c0-9a99-4eb9-8870-a7bfd1d337c8	mari	mari@gmail.com	\$2a\$10\$Mf1kizJAMHV15xA...	voter	true	f638ac71552b3e2a2...	2025-04-12 12:12:05.506+05:30	2025-04-12 12:12:22.624+0
2	1d18b850-6d5a-40c2-9804-e67086eee789	krish	krish@gmail.com	\$2a\$10\$SHDU7WYUw5JslUg...	voter	true	5fff2b974981ba1e8e...	2025-04-13 17:09:34.099+05:30	2025-04-13 17:13:23.074+0
3	4d0ff507-448c-439a-9969-5e766fab0152	deek	deek@gmail.com	\$2a\$10\$N/8UMC5b.8ErCXO...	admin	false	3ad6ce7dceff0503ac...	2025-04-12 12:00:40.888+05:30	2025-04-12 12:00:40.888+0
4	776933c3-328e-4d60-ba76-260a6fb8817a	pavan	pavan@gmail.com	\$2a\$10\$JHyqP.PJzqw6yKH...	voter	true	6443237d2405117a8...	2025-04-12 11:59:46.669+05:30	2025-04-12 12:00:00.61+05
5	9685c433-0186-4cb8-9fa8-6ade883415cd	bro	bro@gmail.com	\$2a\$10\$KkayysMXc2SkVky...	voter	true	37cbaa15a482be461...	2025-04-12 18:17:31.139+05:30	2025-04-12 18:17:44.771+0

Voters Table in DataBase (Postgres)

pgAdmin 4

Welcome public.Users/votin... x public.Votes/voting/postgres@PostgreSQL 17 x

public.Votes/voting/postgres@PostgreSQL 17

Query Query History

```
1 SELECT * FROM public."Votes"
2 ORDER BY id ASC
```

Scratch Pad x

Data Output Messages Notifications

Showing rows: 1 to 4 Page No: 1 of 1

	id [PK] uuid	encryptedVote text	electionid character varying (255)	createdAt timestamp with time zone	updatedAt timestamp with time zone	voterid uuid
1	187c3cf1-3c57-41eb-9c35-00b7a46d71...	U2FsdGVkX18BNo0t9UsXBZ1R0kAb30VcGrUgrRoGm...	1	2025-04-13 17:13:23.061+05:30	2025-04-13 17:13:23.061+05:30	1d18b850-6d5a-40c2-9804-e67086eee7...
2	27dca67d-7c10-4ce6-a64a-958fa71c58f6	U2FsdGVkX1/FD1WeDx6LpnBgJ8Q5L5fNXuqWGX50...	1	2025-04-12 12:12:22.617+05:30	2025-04-12 12:12:22.617+05:30	16bfd7c0-9a99-4eb9-8870-a7bfd1d337c8
3	7ff17b93-7328-45bb-973b-ccd8dc58eb0a	U2FsdGVkX1+ewFc7C0zwa1WjbaGJYr+EI9rMssdCxio=	1	2025-04-12 12:00:00.603+05:30	2025-04-12 12:00:00.603+05:30	776933c3-328e-4d60-ba76-260a6fb881...
4	e7177866-54ce-4431-986e-b090a3f588fd	U2FsdGVkX1+dgSd06KnMQP07VuzYbZx+7hVtb1G4=	1	2025-04-12 18:17:44.766+05:30	2025-04-12 18:17:44.766+05:30	9685c433-0186-4cb8-9fa8-6ade883415cd
